

Assignment #1 part 1, Total points: 40
(Course: CS 401)

These are the first two problems for Assignment 1. **The remaining problems of Assignment 1 will be given out later.**

For regular students, the deadline is **October 5, Monday, in class.**

For special needs students, the deadline is **October 12, Monday, in class.**

Please submit a hard copy of your solution (handwritten or typed) in person in class. No late assignments will be accepted.

Special note: Any answer that is not sufficiently clear even after a reasonably careful reading will not be considered a correct answer, and only what is written in the answer will be used to verify accuracy. No vague descriptions or sufficiently ambiguous statements that can be interpreted in multiple ways will be considered as a correct answer, nor will the student be allowed to add any explanations to his/her answer after it has been submitted.

Problem 1 (20 points): Prof. Smart thinks he is smarter than all the students in this CS 401 class. He has made the following claim to show how smart he is.

Claim made by Prof. Smart: Consider the stable matching problem as taught in class. Suppose that we have only three men, say m_1 , m_2 and m_3 , and only three women, say w_1 , w_2 and w_3 , with their corresponding preference lists. Suppose also that the matching m_1-w_1 , m_2-w_2 , m_3-w_3 is a stable matching. Prof. Smart claims that in that case the matching m_1-w_2 , m_2-w_1 , m_3-w_3 can **never** be a stable matching.

Your task is to decide if Prof. Smart is indeed so smart. For this purpose, do the following.

- Either prove the claim made by Prof. Smart is indeed correct. Such a proof should work **no matter** what the preferences of the men and women are, as long as m_1-w_1 , m_2-w_2 , m_3-w_3 is a stable matching.
- Or, prove the claim made by Prof. Smart is wrong by giving a counter-example. The counter-example should provide the preferences lists of every man and woman, and show that for these preference lists **both** m_1-w_1 , m_2-w_2 , m_3-w_3 and m_1-w_2 , m_2-w_1 , m_3-w_3 are indeed stable matchings.

Problem 2 (20 points): Take the following functions and arrange them in ascending order of growth rates. That is, if function $g(n)$ immediately follows function $f(n)$ in your list, then it

should be the case that $f(n) = O(g(n))$. **Briefly justify your answer.**

$$g_1(n) = 3\sqrt{\log_2 n}$$

$$g_2(n) = 2^n$$

$$g_3(n) = n^{300}$$

$$g_4(n) = n(\log_2 n)^{100}$$

$$g_5(n) = \frac{n^{\log_2 n}}{2345}$$

$$g_6(n) = 2^{2^n}$$

$$g_7(n) = 2^{n^2}$$

If necessary, you may use the following fact that was also discussed in class. If $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)}$ is at most a constant then $f(n) = O(g(n))$.